

02 — GAN 2D Unnormalized

Dự án: Latent Manipulation of Brain MRI using Volume-Preserving GANs

Input: PNG unnormalized từ 00a_preprocessing_2d.ipynb

Kiến trúc:

- **Generator:** U-Net + Age Embedding inject vào bottleneck
- **Discriminator:** PatchGAN + Age Conditioning

Output:

```
gan2d_unnormalized/  
└─ best_model.pth ← checkpoint tốt nhất (loss_G thấp nhất)
```

Bước 1: Cấu hình

```
import os  
  
# ==== ĐƯỜNG DẪN ====  
# Khác file 01: tro' tới unnormalized thay vì normalized  
DATA_DIR =  
'/kaggle/input/datasets/minhbodoi/full-preprocessed-2d/preprocessed_2d/  
unnormalized'  
LABELS_CSV =  
'/kaggle/input/datasets/minhbodoi/full-preprocessed-2d/preprocessed_2d/  
preprocessing_log.csv'  
OUTPUT_DIR = '/kaggle/working/conditional_gan2d_unnormalized'  
  
os.makedirs(OUTPUT_DIR, exist_ok=True)  
  
# ==== HYPERPARAMETERS ====  
IMAGE_SIZE = 256  
BATCH_SIZE = 16  
NUM_EPOCHS = 500 # 2D: 500 epoch  
PATIENCE = 20 # dừng nếu val_SSIM không tăng >= MIN_DELTA sau 20  
epoch liên tiếp  
LR_G = 2e-4  
LR_D = 2e-4  
LAMBDA_L1 = 10 # giảm từ 100 → 10 để model dám thay đổi tuổi  
LAMBDA_AGE = 5 # tăng từ 1 → 5 để age loss có trọng lượng hơn  
LAMBDA_GP = 10 # gradient penalty cho WGAN-GP  
LATENT_DIM = 256  
  
print(f'Data dir : {DATA_DIR}')  
print(f'PNG files: {len([f for f in os.listdir(DATA_DIR) if  
f.endswith(".png")])}')
```

```
Data dir :  
/kaggle/input/datasets/minhbodoi/full-preprocessed-2d/preprocessed_2d/  
unnormalized  
PNG files: 12497
```

Bước 2: Import thư viện

```
import torch  
import torch.nn as nn  
import torch.optim as optim  
from torch.utils.data import Dataset, DataLoader  
import torchvision.transforms as transforms  
import numpy as np  
import pandas as pd  
from PIL import Image  
from tqdm.notebook import tqdm  
  
DEVICE = 'cuda' if torch.cuda.is_available() else 'cpu'  
print(f'Device: {DEVICE}')if DEVICE == 'cuda':  
    print(f'GPU : {torch.cuda.get_device_name(0)}')    print(f'VRAM: {torch.cuda.get_device_properties(0).total_memory /  
1e9:.1f} GB')  
Device: cuda  
GPU : Tesla T4  
VRAM: 15.6 GB
```

Bước 3: Dataset

```
from torch.utils.data import random_split  
  
class BrainMRI2DDataset(Dataset):  
    def __init__(self, data_dir, labels_csv, image_size=256):  
        self.data_dir = data_dir  
        df = pd.read_csv(labels_csv)  
        df['full_path'] = df['png_file'].apply(lambda x:  
os.path.join(data_dir, x))  
        df =  
df[df['full_path'].apply(os.path.exists)].reset_index(drop=True)  
        self.df = df  
        self.age_min = df['age'].min()  
        self.age_max = df['age'].max()  
        self.transform = transforms.Compose([  
            transforms.Resize((image_size, image_size)),  
            transforms.ToTensor(),  
            transforms.Normalize([0.5], [0.5])  
        ])  
        print(f'Dataset: {len(df)} ảnh | tuổi [{self.age_min:.1f},
```

```

{self.age_max:.1f}}')

    def normalize_age(self, age):
        return 2 * (age - self.age_min) / (self.age_max -
self.age_min) - 1

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        row = self.df.iloc[idx]
        img = Image.open(row['full_path']).convert('L')
        img = self.transform(img)
        age_norm = torch.tensor(self.normalize_age(row['age']),
dtype=torch.float32)
        age_raw = torch.tensor(row['age'] / 100.0,
dtype=torch.float32)
        return img, age_norm, age_raw

# — Train / Test split 80/20 —————
full_dataset = BrainMRI2DDataset(DATA_DIR, LABELS_CSV, IMAGE_SIZE)
n_total = len(full_dataset)
n_train = int(0.8 * n_total)
n_test = n_total - n_train

train_set, test_set = random_split(
    full_dataset, [n_train, n_test],
    generator=torch.Generator().manual_seed(42) # seed cố định để
tái lập
)

# Lưu lại age_min/max từ full dataset (không tính riêng train)
dataset = full_dataset # dùng để lấy age_min/max khi save
checkpoint

dataloader = DataLoader(train_set, batch_size=BATCH_SIZE,
                        shuffle=True, num_workers=4,
pin_memory=True)
dataloader_test = DataLoader(test_set, batch_size=BATCH_SIZE,
                        shuffle=False, num_workers=4,
pin_memory=True)

print(f'Train: {n_train} ảnh | Test: {n_test} ảnh')

Dataset: 12497 ảnh | tuổi [18.0, 97.0]
Train: 9997 ảnh | Test: 2500 ảnh

```

Bước 4: Kiến trúc Model

```
class AgeEmbedding(nn.Module):
    def __init__(self, embed_dim=256):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(1, 128), nn.ReLU(),
            nn.Linear(128, embed_dim)
        )
    def forward(self, age):
        return self.net(age.unsqueeze(-1))

class UNetBlock(nn.Module):
    def __init__(self, in_ch, out_ch, down=True, use_bn=True,
        dropout=False):
        super().__init__()
        layers = []
        if down : layers.append(nn.Conv2d(in_ch, out_ch, 4, 2, 1,
            bias=False))
        else : layers.append(nn.ConvTranspose2d(in_ch, out_ch, 4,
            2, 1, bias=False))
        if use_bn : layers.append(nn.BatchNorm2d(out_ch))
        if dropout : layers.append(nn.Dropout(0.5))
        layers.append(nn.LeakyReLU(0.2) if down else nn.ReLU())
        self.block = nn.Sequential(*layers)
    def forward(self, x):
        return self.block(x)

class Generator(nn.Module):
    """
    U-Net Generator với age conditioning.
    Input : ảnh (B, 1, 256, 256) + tuổi normalize (B,)
    Output: ảnh sinh ra (B, 1, 256, 256)
    """
    def __init__(self, latent_dim=256):
        super().__init__()
        self.age_embed = AgeEmbedding(latent_dim)
        self.age_proj = nn.Linear(latent_dim, 512)
        self.age_fuse = nn.Conv2d(1024, 512, 1) # concat e8(512) +
        age(512) → 512
        self.e1 = UNetBlock(1, 64, down=True, use_bn=False)
        self.e2 = UNetBlock(64, 128, down=True)
        self.e3 = UNetBlock(128, 256, down=True)
        self.e4 = UNetBlock(256, 512, down=True)
        self.e5 = UNetBlock(512, 512, down=True)
        self.e6 = UNetBlock(512, 512, down=True)
        self.e7 = UNetBlock(512, 512, down=True)
        self.e8 = UNetBlock(512, 512, down=True, use_bn=False)
```

```

        self.d1 = UNetBlock(512, 512, down=False, dropout=True)
        self.d2 = UNetBlock(1024, 512, down=False, dropout=True)
        self.d3 = UNetBlock(1024, 512, down=False, dropout=True)
        self.d4 = UNetBlock(1024, 512, down=False)
        self.d5 = UNetBlock(1024, 256, down=False)
        self.d6 = UNetBlock(512, 128, down=False)
        self.d7 = UNetBlock(256, 64, down=False)
        self.out = nn.Sequential(nn.ConvTranspose2d(128, 1, 4, 2, 1),
nn.Tanh())

    def forward(self, x, age):
        e1 = self.e1(x); e2 = self.e2(e1); e3 = self.e3(e2); e4 =
self.e4(e3)
        e5 = self.e5(e4); e6 = self.e6(e5); e7 = self.e7(e6); e8 =
self.e8(e7)
        age_feat = self.age_proj(self.age_embed(age)).view(-1, 512, 1,
1).expand_as(e8)
        z = self.age_fuse(torch.cat([e8, age_feat], dim=1)) # concat
→ model tự học balance
        d1 = self.d1(z)
        d2 = self.d2(torch.cat([d1, e7], dim=1))
        d3 = self.d3(torch.cat([d2, e6], dim=1))
        d4 = self.d4(torch.cat([d3, e5], dim=1))
        d5 = self.d5(torch.cat([d4, e4], dim=1))
        d6 = self.d6(torch.cat([d5, e3], dim=1))
        d7 = self.d7(torch.cat([d6, e2], dim=1))
        return self.out(torch.cat([d7, e1], dim=1))

class Discriminator(nn.Module):
    """
    PatchGAN Discriminator với age conditioning.
    Input : ảnh (B, 1, H, W) + age channel → (B, 2, H, W)
    """
    def __init__(self):
        super().__init__()
        self.model = nn.Sequential(
            nn.Conv2d(2, 64, 4, 2, 1, bias=False), nn.LeakyReLU(0.2),
            nn.Conv2d(64, 128, 4, 2, 1, bias=False),
nn.BatchNorm2d(128), nn.LeakyReLU(0.2),
            nn.Conv2d(128, 256, 4, 2, 1, bias=False),
nn.BatchNorm2d(256), nn.LeakyReLU(0.2),
            nn.Conv2d(256, 512, 4, 1, 1, bias=False),
nn.BatchNorm2d(512), nn.LeakyReLU(0.2),
            nn.Conv2d(512, 1, 4, 1, 1)
        )
    def forward(self, img, age):
        age_map = age.view(-1, 1, 1, 1).expand(-1, 1, img.shape[2],
img.shape[3])
        return self.model(torch.cat([img, age_map], dim=1))

```

```

G = Generator(LATENT_DIM).to(DEVICE)
D = Discriminator().to(DEVICE)
print(f'Generator      : {sum(p.numel() for p in
G.parameters())/1e6:.1f}M params')
print(f'Discriminator: {sum(p.numel() for p in
D.parameters())/1e6:.1f}M params')

Generator      : 55.1M params
Discriminator: 2.8M params

```

Bước 5: Loss + Optimizers

Dùng **WGAN-GP** thay vì BCE để tránh mode collapse và training ổn định hơn.

```

# — WGAN-GP: không dùng BCEWithLogitsLoss nữa —————
# Discriminator không dùng sigmoid, output là raw score (Wasserstein)
criterion_l1 = nn.L1Loss()
criterion_age = nn.MSELoss()

def compute_gradient_penalty(D, real, fake, ages, device):
    """Gradient penalty để ổn định training WGAN-GP."""
    B = real.size(0)
    alpha = torch.rand(B, 1, 1, 1, device=device)
    interp = (alpha * real + (1 - alpha) * fake).requires_grad_(True)
    d_interp = D(interp, ages)
    grad = torch.autograd.grad(
        outputs=d_interp, inputs=interp,
        grad_outputs=torch.ones_like(d_interp),
        create_graph=True, retain_graph=True
    )[0]
    gp = ((grad.norm(2, dim=1) - 1) ** 2).mean()
    return gp

age_regressor = nn.Sequential(
    nn.AdaptiveAvgPool2d(8), nn.Flatten(),
    nn.Linear(64, 64), nn.ReLU(),
    nn.Linear(64, 1), nn.Sigmoid()
).to(DEVICE)

opt_G = optim.Adam(
    list(G.parameters()) + list(age_regressor.parameters()),
    lr=LR_G, betas=(0.0, 0.9) # betas WGAN-GP chuẩn
)
opt_D = optim.Adam(D.parameters(), lr=LR_D, betas=(0.0, 0.9))

scheduler_G = optim.lr_scheduler.StepLR(opt_G, step_size=50,
gamma=0.5)

```

```
scheduler_D = optim.lr_scheduler.StepLR(opt_D, step_size=50,
gamma=0.5)
```

```
print('Loss + Optimizers (WGAN-GP) sẵn sàng!')
```

```
Loss + Optimizers (WGAN-GP) sẵn sàng!
```

Bước 6: Training

```
from skimage.metrics import structural_similarity as ssim_fn
import numpy as np, subprocess, shutil, tempfile

# — Duong dan checkpoint —————
CKPT_PATH    = f'{OUTPUT_DIR}/last_checkpoint.pth'
BEST_PATH    = f'{OUTPUT_DIR}/best_model.pth'

# — Kaggle Dataset info —————
_ku = subprocess.run('kaggle config view', shell=True,
capture_output=True, text=True).stdout
KAGGLE_USER  = [l.split(':')[1].strip() for l in _ku.split('\n') if
'username' in l][0]
DATASET_NAME = os.path.basename(OUTPUT_DIR).replace('_', '-')
MIN_DELTA    = 0.005 # chỉ tính là 'cải thiện thật' khi SSIM tăng >=
0.005

def push_checkpoints_to_kaggle(msg):
    tmp = tempfile.mkdtemp()
    try:
        for fn in ['last_checkpoint.pth', 'best_model.pth']:
            if os.path.exists(f'{OUTPUT_DIR}/{fn}'):
                shutil.copy2(f'{OUTPUT_DIR}/{fn}', f'{tmp}/{fn}')
        import json as _j
        with open(f'{tmp}/dataset-metadata.json', 'w') as _f:
            _j.dump({'title': DATASET_NAME,
                    'id'   : f'{KAGGLE_USER}/{DATASET_NAME}',
                    'licenses': [{ 'name': 'CC0-1.0' }]}, _f)
        chk = subprocess.run(
            f'kaggle datasets list --user {KAGGLE_USER} --search
{DATASET_NAME}',
            shell=True, capture_output=True, text=True)
        if DATASET_NAME in chk.stdout:
            subprocess.run(f'kaggle datasets version -p {tmp} -m
"{msg}"', shell=True)
        else:
            subprocess.run(f'kaggle datasets create -p {tmp}',
shell=True)
        print(f' Cloud: {msg} -> {KAGGLE_USER}/{DATASET_NAME}')
    finally:
        shutil.rmtree(tmp)
```

```

# — Resume neu co checkpoint —————
start_epoch      = 1
patience_count  = 0
best_val_ssim    = -1.0
history = {'loss_G': [], 'loss_D': [], 'loss_L1': [], 'loss_age': [],
          'val_ssim': []}

def find_checkpoint(filename):
    p = f'{OUTPUT_DIR}/{filename}'
    if os.path.exists(p): return p, 'working'
    import glob
    matches = glob.glob(f'/kaggle/input/**/{filename}',
recursive=True)
    if matches: return matches[0], 'input'
    return None, None

load_path = None
for fname in ['last_checkpoint.pth', 'best_model.pth']:
    p, src_loc = find_checkpoint(fname)
    if p:
        load_path = p
        print(f'Tim thay {fname} ({src_loc}) — load de train tiep...')
        break
if not load_path:
    print('Khong co checkpoint — train tu dau')

if load_path:
    ckpt = torch.load(load_path, map_location=DEVICE)
    G.load_state_dict(ckpt['G_state'])
    D.load_state_dict(ckpt['D_state'])
    opt_G.load_state_dict(ckpt['opt_G'])
    opt_D.load_state_dict(ckpt['opt_D'])
    start_epoch      = ckpt['epoch'] + 1
    best_val_ssim     = ckpt.get('best_val_ssim', -1.0)
    patience_count    = ckpt.get('patience_count', 0)
    history           = ckpt.get('history', history)
    print(f'Resume tu epoch {ckpt["epoch"]} |
best_val_SSIM={best_val_ssim:.4f}')
    print(f'    Tiep tuc tu epoch {start_epoch} -> {NUM_EPOCHS}')

N_CRITIC = 5
print(f'Bat dau training {NUM_EPOCHS} epochs (WGAN-GP)...\n')

for epoch in range(start_epoch, NUM_EPOCHS + 1):
    G.train(); D.train()
    epoch_loss_G = epoch_loss_D = epoch_loss_L1 = epoch_loss_age = 0
    n_batches = 0

    data_iter = iter(dataloader)
    for _ in range(len(dataloader) // N_CRITIC):

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```

for _ in range(N_CRITIC):
    try:
        real_imgs, ages_norm, ages_raw = next(data_iter)
    except StopIteration:
        data_iter = iter(dataloader)
        real_imgs, ages_norm, ages_raw = next(data_iter)
    real_imgs = real_imgs.to(DEVICE)
    ages_norm = ages_norm.to(DEVICE)
    ages_raw = ages_raw.to(DEVICE)
    opt_D.zero_grad()
    with torch.no_grad():
        fake_imgs = G(real_imgs, ages_norm)
        loss_D_real = -D(real_imgs, ages_norm).mean()
        loss_D_fake = D(fake_imgs.detach(), ages_norm).mean()
        gp = compute_gradient_penalty(D, real_imgs,
fake_imgs.detach(), ages_norm, DEVICE)
        loss_D = loss_D_real + loss_D_fake + LAMBDA_GP * gp
        loss_D.backward()
        opt_D.step()
        epoch_loss_D += loss_D.item()

    try:
        real_imgs, ages_norm, ages_raw = next(data_iter)
    except StopIteration:
        data_iter = iter(dataloader)
        real_imgs, ages_norm, ages_raw = next(data_iter)
    real_imgs = real_imgs.to(DEVICE)
    ages_norm = ages_norm.to(DEVICE)
    ages_raw = ages_raw.to(DEVICE)
    opt_G.zero_grad()
    fake_imgs = G(real_imgs, ages_norm)
    loss_G_adv = -D(fake_imgs, ages_norm).mean()
    loss_L1 = criterion_l1(fake_imgs, real_imgs) * LAMBDA_L1
    pred_age = age_regressor(fake_imgs).squeeze()
    loss_age = criterion_age(pred_age, ages_raw) * LAMBDA_AGE
    loss_G = loss_G_adv + loss_L1 + loss_age
    loss_G.backward()
    opt_G.step()
    epoch_loss_G += loss_G_adv.item()
    epoch_loss_L1 += loss_L1.item()
    epoch_loss_age += loss_age.item()
    n_batches += 1

scheduler_G.step()
scheduler_D.step()

n = max(n_batches, 1)
avg_loss_G = epoch_loss_G / n
avg_loss_D = epoch_loss_D / (n * N_CRITIC)
avg_loss_L1 = epoch_loss_L1 / n

```

```

avg_loss_age = epoch_loss_age / n

G.eval()
ssim_scores = []
with torch.no_grad():
    for real_imgs, ages_norm, _ in dataloader_test:
        real_imgs = real_imgs.to(DEVICE)
        shuffled_idx = torch.randperm(len(ages_norm))
        ages_shuffled = ages_norm[shuffled_idx].to(DEVICE)
        fake_imgs = G(real_imgs, ages_shuffled)
        for i in range(real_imgs.size(0)):
            r = (real_imgs[i, 0].cpu().numpy() + 1) / 2
            f = (fake_imgs[i, 0].cpu().numpy() + 1) / 2
            ssim_scores.append(ssim_fn(r, f, data_range=1.0))
val_ssim = float(np.mean(ssim_scores))

history['loss_G'].append(avg_loss_G)
history['loss_D'].append(avg_loss_D)
history['loss_L1'].append(avg_loss_L1)
history['loss_age'].append(avg_loss_age)
history['val_ssim'].append(val_ssim)

# — Luu last_checkpoint —————
torch.save({
    'epoch' : epoch,
    'G_state' : G.state_dict(),
    'D_state' : D.state_dict(),
    'opt_G' : opt_G.state_dict(),
    'opt_D' : opt_D.state_dict(),
    'history' : history,
    'age_min' : dataset.age_min,
    'age_max' : dataset.age_max,
    'best_val_ssim' : best_val_ssim,
    'patience_count' : patience_count,
    'test_indices' : test_set.indices,
}, CKPT_PATH)

# — Luu best model neu cai thien —————
if val_ssim > best_val_ssim + MIN_DELTA:
    best_val_ssim = val_ssim
    patience_count = 0
    torch.save({
        'epoch' : epoch,
        'G_state' : G.state_dict(),
        'D_state' : D.state_dict(),
        'opt_G' : opt_G.state_dict(),
        'opt_D' : opt_D.state_dict(),
        'history' : history,
        'age_min' : dataset.age_min,
        'age_max' : dataset.age_max,

```

```

        'best_loss_G' : avg_loss_G,
        'best_val_ssim' : best_val_ssim,
        'val_ssim' : val_ssim,
        'test_indices' : test_set.indices,
    }, BEST_PATH)
    print(f'Epoch {epoch:3d}/{NUM_EPOCHS} | '
          f'loss_G={avg_loss_G:.4f} | loss_D={avg_loss_D:.4f} | '
          f'loss_L1={avg_loss_L1:.4f} | val_SSIM={val_ssim:.4f} | '
[BEST]')
    else:
        patience_count += 1
        print(f'Epoch {epoch:3d}/{NUM_EPOCHS} | '
              f'loss_G={avg_loss_G:.4f} | loss_D={avg_loss_D:.4f} | '
              f'loss_L1={avg_loss_L1:.4f} | val_SSIM={val_ssim:.4f} | '
              f'no improve {patience_count}/{PATIENCE}')

# — Push moi epoch —————
push_checkpoints_to_kaggle(f'{epoch}/{NUM_EPOCHS}')

# — Early stopping —————
if patience_count >= PATIENCE:
    print(f'Early stopping tai epoch {epoch}')
    break

print(f'\n=== TRAINING HOAN THANH ===')
print(f'Best val_SSIM: {best_val_ssim:.4f}')
print(f'Model luu tai: {OUTPUT_DIR}/best_model.pth')

```

Khong co checkpoint – train tu dau
 Bat dau training 500 epochs (WGAN-GP)...

```
Epoch 1/500 | loss_G=-1.0173 | loss_D=0.3232 | loss_L1=1.0931 |
val_SSIM=0.8857 | [BEST]
```

Warning: Looks like you're using an outdated `kaggle` version
 (installed: 2.0.1), please consider upgrading to the latest version
 (2.0.2)

Starting upload for file best_model.pth

```
100%|██████████| 662M/662M [00:05<00:00, 132MB/s]
 0%|          | 0.00/662M [00:00<?, ?B/s]
```

Upload successful: best_model.pth (662MB)
 Starting upload for file last_checkpoint.pth

```
100%|██████████| 662M/662M [00:08<00:00, 79.1MB/s]
```

Upload successful: last_checkpoint.pth (662MB)
 Your private Dataset is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
 Cloud: 1/500 -> dyiol47/conditional-gan2d-unnormalized

Epoch 2/500 | loss_G=2.5293 | loss_D=1.5011 | loss_L1=0.3863 |
val_SSIM=0.9248 | [BEST]
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 129MB/s]
0%|██████████| 0.00/662M [00:00<?, ?B/s]

Upload successful: best_model.pth (662MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:05<00:00, 125MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 2/500 -> dyiol47/conditional-gan2d-unnormalized

Epoch 3/500 | loss_G=3.8813 | loss_D=1.5589 | loss_L1=0.3142 |
val_SSIM=0.9360 | [BEST]

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:08<00:00, 78.2MB/s]

Upload successful: best_model.pth (662MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:05<00:00, 138MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 3/500 -> dyiol47/conditional-gan2d-unnormalized

Epoch 4/500 | loss_G=1.5268 | loss_D=1.6579 | loss_L1=0.3232 |
val_SSIM=0.9318 | no improve 1/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 120MB/s]
0%|██████████| 0.00/662M [00:00<?, ?B/s]

Upload successful: best_model.pth (662MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:06<00:00, 111MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 4/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 5/500 | loss_G=-2.1282 | loss_D=1.4921 | loss_L1=0.2825 |
val_SSIM=0.9339 | no improve 2/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 124MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

Upload successful: best_model.pth (662MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:06<00:00, 110MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 5/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 6/500 | loss_G=-3.6462 | loss_D=1.7699 | loss_L1=0.2864 |
val_SSIM=0.9416 | [BEST]
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 132MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:09<00:00, 75.4MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 6/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 7/500 | loss_G=-7.3489 | loss_D=1.7107 | loss_L1=0.2905 |
val_SSIM=0.9380 | no improve 1/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:07<00:00, 91.8MB/s]
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100%|██████████| 662M/662M [00:08<00:00, 86.8MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 7/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 8/500 | loss_G=-9.9803 | loss_D=2.0260 | loss_L1=0.2931 |
val_SSIM=0.9343 | no improve 2/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:06<00:00, 107MB/s]
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100%|██████████| 662M/662M [00:12<00:00, 53.8MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 8/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 9/500 | loss_G=-12.4961 | loss_D=2.0646 | loss_L1=0.2979 |
val_SSIM=0.9443 | no improve 3/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 126MB/s]

Upload successful: best_model.pth (662MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:07<00:00, 88.0MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 9/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 10/500 | loss_G=-11.2216 | loss_D=1.8327 | loss_L1=0.2790 |
val_SSIM=0.7783 | no improve 4/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version

(2.0.2)

Starting upload for file best_model.pth

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100%|██████████| 662M/662M [00:05<00:00, 128MB/s]
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100%|██████████| 662M/662M [00:05<00:00, 127MB/s]
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Dataset version is being created. Please check progress at

<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 10/500 -> dyiol47/conditional-gan2d-unnormalized

Epoch 11/500 | loss_G=-11.2792 | loss_D=1.8741 | loss_L1=0.2704 |
val_SSIM=0.9343 | no improve 5/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

```
100%|██████████| 662M/662M [00:09<00:00, 70.1MB/s]
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Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at

<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 11/500 -> dyiol47/conditional-gan2d-unnormalized

Epoch 12/500 | loss_G=-10.3407 | loss_D=1.6133 | loss_L1=0.2220 |
val_SSIM=0.9686 | [BEST]

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

```
100%|██████████| 662M/662M [00:08<00:00, 85.7MB/s]
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```
100%|██████████| 662M/662M [00:10<00:00, 69.4MB/s]
```

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Dataset version is being created. Please check progress at

<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 12/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 13/500 | loss_G=-9.6072 | loss_D=1.5205 | loss_L1=0.2295 |
val_SSIM=0.9528 | no improve 1/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:11<00:00, 62.8MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:07<00:00, 89.0MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 13/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 14/500 | loss_G=-11.0104 | loss_D=1.4590 | loss_L1=0.2325 |
val_SSIM=0.9611 | no improve 2/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:07<00:00, 88.1MB/s]
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100%|██████████| 662M/662M [00:09<00:00, 72.4MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 14/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 15/500 | loss_G=-15.6828 | loss_D=1.6594 | loss_L1=0.2506 |
val_SSIM=0.9663 | no improve 3/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:08<00:00, 78.0MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:07<00:00, 92.1MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 15/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 16/500 | loss_G=-11.8463 | loss_D=1.5436 | loss_L1=0.2290 |
val_SSIM=0.9544 | no improve 4/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 129MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:10<00:00, 64.9MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 16/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 17/500 | loss_G=-9.9978 | loss_D=1.4231 | loss_L1=0.2033 |
val_SSIM=0.9574 | no improve 5/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
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100%|██████████| 662M/662M [00:08<00:00, 86.0MB/s]
0%| | 1.33M/662M [00:00<00:54, 12.7MB/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:07<00:00, 92.0MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 17/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 18/500 | loss_G=-8.2054 | loss_D=1.3959 | loss_L1=0.2167 |
val_SSIM=0.9665 | no improve 6/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 129MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:05<00:00, 122MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 18/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 19/500 | loss_G=-6.7565 | loss_D=1.3237 | loss_L1=0.2060 |
val_SSIM=0.9660 | no improve 7/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:07<00:00, 96.3MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:07<00:00, 88.5MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 19/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 20/500 | loss_G=-6.6056 | loss_D=1.1395 | loss_L1=0.1895 |
val_SSIM=0.9677 | no improve 8/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 121MB/s]
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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:07<00:00, 89.6MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 20/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 21/500 | loss_G=-6.0428 | loss_D=0.9601 | loss_L1=0.1790 |
val_SSIM=0.9674 | no improve 9/20

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version (2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 121MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:06<00:00, 101MB/s]

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Dataset version is being created. Please check progress at

<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 21/500 -> dyiol47/conditional-gan2d-unnormalized

Epoch 22/500 | loss_G=-4.4170 | loss_D=0.8784 | loss_L1=0.1736 |

val_SSIM=0.9751 | [BEST]

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version (2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:06<00:00, 106MB/s]
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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:08<00:00, 82.5MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at

<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 22/500 -> dyiol47/conditional-gan2d-unnormalized

Epoch 23/500 | loss_G=-4.3097 | loss_D=0.8208 | loss_L1=0.1703 |

val_SSIM=0.9723 | no improve 1/20

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version (2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:09<00:00, 75.2MB/s]

Upload successful: best_model.pth (662MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:08<00:00, 83.2MB/s]

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Dataset version is being created. Please check progress at

<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 23/500 -> dyiol47/conditional-gan2d-unnormalized

Epoch 24/500 | loss_G=-4.3391 | loss_D=0.8133 | loss_L1=0.1705 |

val_SSIM=0.9634 | no improve 2/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:08<00:00, 83.1MB/s]

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Upload successful: best_model.pth (662MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:06<00:00, 106MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at

<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 24/500 -> dyiol47/conditional-gan2d-unnormalized

Epoch 25/500 | loss_G=-4.7142 | loss_D=0.7743 | loss_L1=0.1595 |

val_SSIM=0.9703 | no improve 3/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:06<00:00, 104MB/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:05<00:00, 117MB/s]

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Dataset version is being created. Please check progress at

<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 25/500 -> dyiol47/conditional-gan2d-unnormalized

Epoch 26/500 | loss_G=-4.1523 | loss_D=0.7475 | loss_L1=0.1627 |

val_SSIM=0.9763 | no improve 4/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 134MB/s]

0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:05<00:00, 128MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 26/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 27/500 | loss_G=-3.6975 | loss_D=0.7405 | loss_L1=0.1587 |
val_SSIM=0.9747 | no improve 5/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:06<00:00, 100MB/s]

Upload successful: best_model.pth (662MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:05<00:00, 136MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 27/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 28/500 | loss_G=-3.9863 | loss_D=0.7364 | loss_L1=0.1579 |
val_SSIM=0.9765 | no improve 6/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:06<00:00, 113MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

Upload successful: best_model.pth (662MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:05<00:00, 122MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 28/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 29/500 | loss_G=-3.8586 | loss_D=0.6872 | loss_L1=0.1540 |
val_SSIM=0.9748 | no improve 7/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version

(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 118MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

Upload successful: best_model.pth (662MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:05<00:00, 129MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at

<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 29/500 -> dyiol47/conditional-gan2d-unnormalized

Epoch 30/500 | loss_G=-3.4580 | loss_D=0.6640 | loss_L1=0.1593 |
val_SSIM=0.9697 | no improve 8/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 121MB/s]
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100%|██████████| 662M/662M [00:10<00:00, 65.9MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at

<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 30/500 -> dyiol47/conditional-gan2d-unnormalized

Epoch 31/500 | loss_G=-4.1656 | loss_D=0.6489 | loss_L1=0.1629 |
val_SSIM=0.9767 | no improve 9/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:08<00:00, 82.2MB/s]
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100%|██████████| 662M/662M [00:09<00:00, 74.9MB/s]

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Dataset version is being created. Please check progress at

<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 31/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 32/500 | loss_G=-4.4181 | loss_D=0.6235 | loss_L1=0.1694 |
val_SSIM=0.9704 | no improve 10/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:07<00:00, 88.7MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:07<00:00, 96.9MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 32/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 33/500 | loss_G=-4.8496 | loss_D=0.6017 | loss_L1=0.1670 |
val_SSIM=0.9771 | no improve 11/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 123MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:09<00:00, 74.3MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 33/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 34/500 | loss_G=-4.9502 | loss_D=0.5898 | loss_L1=0.1665 |
val_SSIM=0.9754 | no improve 12/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:07<00:00, 99.2MB/s]
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100%|██████████| 662M/662M [00:08<00:00, 80.4MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 34/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 35/500 | loss_G=-4.5026 | loss_D=0.5582 | loss_L1=0.1757 |
val_SSIM=0.9752 | no improve 13/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 130MB/s]
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100%|██████████| 662M/662M [00:07<00:00, 93.2MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 35/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 36/500 | loss_G=-4.8370 | loss_D=0.5639 | loss_L1=0.1689 |
val_SSIM=0.9724 | no improve 14/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 121MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:08<00:00, 78.7MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 36/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 37/500 | loss_G=-4.4849 | loss_D=0.5444 | loss_L1=0.1664 |
val_SSIM=0.9761 | no improve 15/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:07<00:00, 93.8MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:05<00:00, 131MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 37/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 38/500 | loss_G=-4.1084 | loss_D=0.5527 | loss_L1=0.1668 |
val_SSIM=0.9753 | no improve 16/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:11<00:00, 59.4MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:06<00:00, 107MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 38/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 39/500 | loss_G=-3.9046 | loss_D=0.5332 | loss_L1=0.1664 |
val_SSIM=0.9744 | no improve 17/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 132MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

Upload successful: best_model.pth (662MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:06<00:00, 111MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 39/500 -> dyiol47/conditional-gan2d-unnormalized
Epoch 40/500 | loss_G=-4.1417 | loss_D=0.5437 | loss_L1=0.1720 |
val_SSIM=0.9716 | no improve 18/20

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version (2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:06<00:00, 105MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

Upload successful: best_model.pth (662MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:07<00:00, 90.1MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at

<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 40/500 -> dyiol47/conditional-gan2d-unnormalized

Epoch 41/500 | loss_G=-4.0855 | loss_D=0.5348 | loss_L1=0.1629 |

val_SSIM=0.9749 | no improve 19/20

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version (2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:11<00:00, 62.8MB/s]
0%| | 2.86M/662M [00:00<00:24, 28.6MB/s]

Upload successful: best_model.pth (662MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:09<00:00, 76.5MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at

<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>

Cloud: 41/500 -> dyiol47/conditional-gan2d-unnormalized

Epoch 42/500 | loss_G=-3.8608 | loss_D=0.5336 | loss_L1=0.1622 |

val_SSIM=0.9681 | no improve 20/20

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version (2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:08<00:00, 86.4MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

Upload successful: best_model.pth (662MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:05<00:00, 124MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/dyiol47/conditional-gan2d-unnormalized>
Cloud: 42/500 -> dyiol47/conditional-gan2d-unnormalized
Early stopping tại epoch 42

=== TRAINING HOAN THANH ===

Best val_SSIM: 0.9751

Model lưu tại:

/kaggle/working/conditional_gan2d_unnormalized/best_model.pth

Push đã được tích hợp vào training loop (cell trên) – chạy mỗi epoch.

print(f'Checkpoints đã được push lên: {KAGGLE_USER}/{DATASET_NAME}')

Checkpoints đã được push lên: dyiol47/conditional-gan2d-unnormalized